SDG 7 AFFORDABLE AND CLEAN ENERGY PROGRESS REPORT

Sustainable Development Goal 7 (SDG 7): Affordable and Clean Energy aims to ensure access to affordable, reliable, sustainable, and modern energy for all. This goal emphasizes the importance of expanding the use of renewable energy, improving energy efficiency, and promoting sustainable energy practices across different sectors of society. For educational institutions such as the City College of Cagayan de Oro (CCCDO), SDG 7 provides a framework for integrating clean energy initiatives into campus operations, infrastructure development, research, and student learning. Through institutional policies, conservation programs, and partnerships with external organizations, colleges can contribute to reducing energy waste and encouraging greater awareness of sustainable energy use within the academic community.

This comparative report examines CCCDO's documented progress toward SDG 7 in 2024 and 2025, focusing on the College's sustainability initiatives, energy conservation efforts, institutional partnerships, and renewable energy projects. The 2024 documentation presents a broad range of institutional activities, including renewable-energy planning, campus energy and water monitoring, conservation campaigns, policy development, inter-agency collaboration, and the integration of sustainability into research and curriculum development. One notable milestone was the exploratory meeting with Greenergy Solar PH on April 23, 2024, which aimed to discuss potential collaboration on sustainability initiatives. This engagement reflected the College's interest in exploring renewable energy opportunities and strengthening partnerships that could support its long-term environmental objectives (CCCDO, 2024).

In 2025, the documented initiatives show further development in the College's engagement with renewable energy, particularly through activities associated with the Agusan Campus. The supplied narrative describes an ocular visit conducted by Mindanao Goes Solar in March 2025, which provided an opportunity to assess the campus and explore the potential application of solar energy. This development was followed by the formal turnover of a solar-powered study shed on June 2, 2025, and the signing of the Deed of Donation on June 7, 2025, as confirmed by CCCDO's official article. The facility was presented as a sustainable learning space where students could study, collaborate, and charge their electronic devices using solar-powered electricity. Beyond providing a practical application of renewable energy, the project also demonstrated how clean energy infrastructure can support students' academic activities while promoting environmental responsibility within the campus community (CCCDO, 2025).

The comparison of two years provides an opportunity to understand how CCCDO's documented SDG 7 initiatives have developed over time. While the 2024 records emphasize planning, monitoring, conservation, policy development, and the exploration of institutional partnerships, the 2025 documentation highlights a more tangible implementation milestone through the establishment of a solar-powered learning facility. This progression suggests a movement from broad institutional discussions and preparatory activities toward a visible

application of renewable energy in the campus environment. It also illustrates the potential value of collaboration between educational institutions and external organizations in advancing sustainability initiatives.

Nevertheless, evaluating progress toward affordable and clean energy requires more than identifying completed activities or newly established facilities. Reliable and comparable indicators are necessary to determine whether these initiatives have resulted in measurable improvements in energy efficiency, reduced electricity costs, increased renewable energy generation, or lower greenhouse gas emissions. Although the 2024 report provides a distribution of SDG 7 program areas and selected indicators, the reviewed 2025 documentation does not provide a common, verified dataset covering electricity consumption, solar energy generation, financial savings, avoided emissions, or the continued implementation of campus-wide conservation measures. Consequently, this report assesses CCCDO's progress based on documented programs, institutional developments, partnerships, and implementation milestones rather than claiming a specific percentage increase in energy performance. This approach provides a more balanced assessment of the College's contributions to SDG 7 while identifying the importance of consistent monitoring, measurable outcomes, and continued investment in sustainable energy initiatives.

Key Findings

- The 2024 SDG 7 report presents five program areas: renewable-energy initiatives, digital monitoring, conservation campaigns, policy and partnerships, and research and education.
- The report attributes the largest share of its 2024 contribution distribution to renewable-energy installation (30.8%), followed by digital monitoring and ICT systems (23.1%) and water/power conservation campaigns (19.2%).
- The April 2024 Greenergy Solar PH engagement is described by CCCDO as an exploratory meeting to discuss potential areas of partnership, not as evidence that a campus-wide solar system was already installed.
- The 2025 record adds a site-level solar assessment narrative and a completed solar-powered study shed formally turned over in June 2025.
- The study shed is a tangible clean-energy and student-learning milestone, but available sources do not quantify its electricity generation, cost savings, emissions reduction, or number of unique users.
- The 2024 institutional progress PDF includes the solar study shed in its summary of achievements, while the official news story dates formal turnover to June 2, 2025. The comparison therefore uses the dated public event for chronology and labels the distribution figure as the report's own attribution.

Institutional Context and SDG 7

SDG 7 aims to ensure access to affordable, reliable, sustainable, and modern energy. For a higher education institution, this goal intersects with the reliability of learning spaces, operational efficiency, renewable-energy adoption, environmental education, research, and cooperation with technology providers and public agencies. Energy is not only a utility cost: it supports lighting, digital learning, access to devices, laboratories, and the daily operation of academic and administrative services.

CCCDO's 2024 progress report connects clean energy with Republic Act No. 11285, the Energy Efficiency and Conservation Act, and describes a combination of infrastructure, monitoring, behavior-change, policy, and educational measures. Its program portfolio includes an ICT-based electricity and water monitoring system, the Switch-Off Drive, Green Classroom Challenge, Eco-Literacy Month, a localized version of energy-efficiency guidelines, collaborations with the City Engineering Office, Cagayan de Oro Water District (COWD), and DOE-10, along with research and curriculum activities (CCCDO, n.d.).

The 2025 dated initiatives complement that institutional framework by showing clean energy in a student-facing setting. The solar-powered study shed demonstrates how a renewable-energy technology can become part of campus life and provide a visible, practical point of reference for sustainability learning. It should be understood as one implemented facility within a broader energy agenda, rather than as evidence of institution-wide energy independence.

2024 Baseline: Policy, Monitoring, Conservation, and Partnership Exploration

Renewable-energy agenda and Greenergy Solar PH

On April 23, 2024, CCCDO's International Linkages and Promotions Office held an exploratory meeting with Greenergy Solar PH. The official article, published May 6, 2024, describes discussions about potential areas of partnership and a shared interest in a greener and more sustainable city. This meeting is important as an early partnership-development step: it placed renewable energy within the College's institutional planning and created a channel for exploring possible collaboration with an energy provider (CCCDO, 2024).

The distinction between exploration and implementation matters. The official report characterizes the engagement as a discussion of potential areas of partnership. It does not itself confirm a signed implementation agreement, commissioned solar array, installed capacity, or electricity savings arising from the meeting. In this comparative analysis, the 2024 event is treated as the strategic and partnership-building baseline, not as proof of completed solar infrastructure.

Digital energy monitoring and campus management

The SDG 7 progress PDF describes an ICT-based digital monitoring system designed to track campus electricity and water use in real time. Its 2024 summary associates the monitoring category with three buildings. In principle, building-level monitoring can make energy use more visible, help facilities teams identify consumption patterns, and support operational decisions about lighting, equipment, and other loads. The source presents the system as part of a data-driven approach to resource management, although it does not provide time-series consumption values or demonstrate a measured reduction in kilowatt-hours.

Conservation campaigns and energy literacy

The report names the Switch-Off Drive, Green Classroom Challenge, and Eco-Literacy Month as campaigns intended to build habits of responsible energy and water use among students and faculty. Its summary associates the conservation-campaign category with a reported reach of 2,000 students and faculty. These initiatives treat energy efficiency as both an operational and educational issue: turning off unnecessary loads and making consumption visible can reinforce everyday conservation behavior across the campus community.

The 2024 report also places environmental researchers and curriculum work within the SDG 7 portfolio. By linking environmental ethics to practical campus action, awareness campaigns offer students opportunities to relate sustainability concepts to routines in classrooms and shared facilities. The reported reach indicates the scale recorded by the source; it does not establish how many participants changed behavior or how much electricity was saved.

Policy development, partnerships, research, and curriculum

The 2024 report describes policy work and partnerships involving the City Engineering Office, COWD, and DOE-10, as well as energy audits and joint activities related to LED retrofitting and building-design optimization. The report's contribution table lists four inter-agency collaborations under policy development and partnerships. The report also describes research on solar integration economics and the social and behavioral factors affecting energy use in higher education, along with energy literacy integrated into STEM and General Education courses.

These components matter because renewable energy adoption is rarely only a matter of buying equipment. It also involves rules, procurement and technical coordination, building performance, user behavior, knowledge development, and institutional capacity. The 2024 portfolio, as described in the source report, presents clean energy as a cross-functional objective connecting facilities management, partnerships, education, and research.

2024 Contribution Distribution: Figure and Interpretation

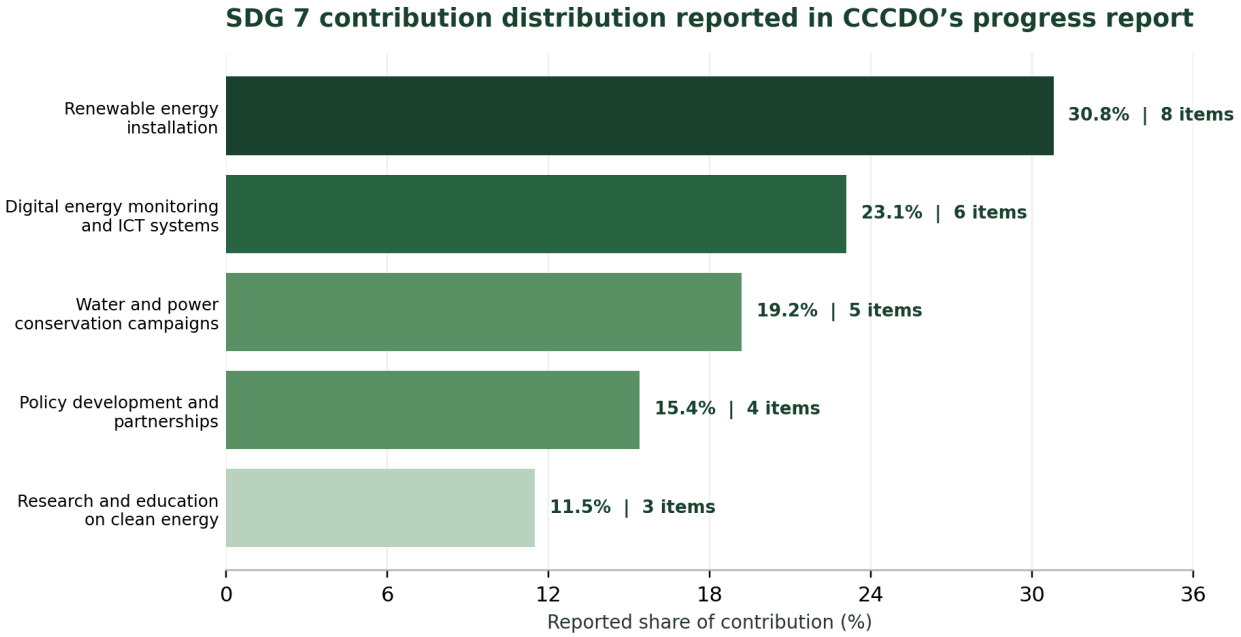


Figure 1. Percentage distribution of SDG 7 contribution categories reproduced from the supplied institutional progress report. Counts shown are the figures in the report's chart.

Interpretation: Renewable-energy installation is the largest category in the report's distribution at 30.8%, followed by digital monitoring and ICT systems at 23.1%, and water/power conservation campaigns at 19.2%. Policy development and partnerships account for 15.4%, while research and clean-energy education account for 11.5%. Read as a reported portfolio profile, the distribution indicates that infrastructure and monitoring were prominent, with

conservation, cooperation, and education acting as complementary strands rather than stand-alone efforts.

The figures are based on the source report's chart, which displays counts of 8, 6, 5, 4, and 3 items respectively, totaling 26. The table in the same PDF separately names five research papers under the research-and-education category, so that "five papers" should not be confused with the chart's count of three category items. The chart percentages are retained as reported rather than recast as energy savings or program effectiveness.

2025 Progress: From Assessment to a Visible Solar Learning Facility

Mindanao Goes Solar ocular visit

The supplied 2025 narrative describes an ocular visit by representatives of the Mindanao Goes Solar program at CCCDO's Agusan Campus in March 2025. The stated purpose was to assess the site's solar potential, identify suitable installation locations, and consider how solar infrastructure might support administrative and educational facilities. This marks a practical next step after the 2024 exploratory discussion with Greenergy Solar PH: attention shifted from general partnership possibilities to a site-focused assessment with another clean-energy initiative.

The ocular visit is evidence of assessment and project preparation rather than a quantified energy outcome. The narrative does not specify the assessment's technical findings, proposed kilowatt capacity, expected annual generation, procurement or installation cost, or the number of buildings selected for solarization. Its contribution to the progress story is that site feasibility and future deployment were explicitly considered in 2025.

Solar-powered study shed: commissioning and turnover

CCCDO's official article reports that the solar-powered study shed was formally turned over on June 2, 2025. The facility was received by the College President and Central Student Government President on behalf of the campus community. A Deed of Donation was ceremonially signed on June 7, 2025, during the Solar Expo at Ayala Malls Centrio Activity Center. The project leads named in the official account were Ms. Magi Caroline Ann Mendoza, Project Supervisor, and Ms. Helen Grace Uy, Managing Director of Mindanao Goes Solar (CCCDO, 2025).

The study shed is the clearest implementation milestone in the 2025 record because it is a completed and identifiable facility, not only a policy or exploratory conversation. The official article describes it as a safe, sustainable space for students to study, collaborate, and power their devices. Its value is therefore both infrastructural and educational: it brings a renewable-energy application into a student-used setting and makes clean-energy technology visible within campus life.

The study shed also provides a concrete link between SDG 7 and the College's educational mission. A solar-powered space can help students understand how renewable electricity is used in practice; it can make energy infrastructure part of everyday learning rather than an abstract sustainability topic. The available source, however, does not quantify the shed's rated capacity, actual kilowatt-hour generation, electricity avoided from the grid, maintenance cost, or total unique users. These benefits are accordingly described in terms of documented function and intended contribution rather than numerical impact.

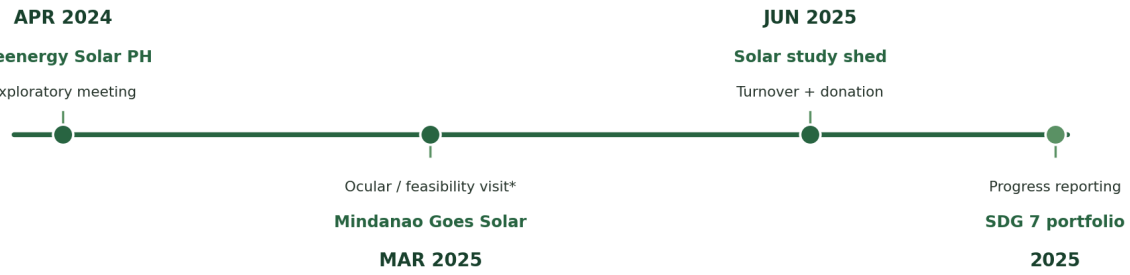
The 2025 SDG 7 portfolio and institutional continuity

The supplied 2025 narrative states that CCCDO compiled an ongoing progress report around solar-powered facilities, energy-consumption reduction, cost considerations, curriculum integration, and community awareness. The related SDG 7 PDF describes a wider portfolio that includes digital monitoring, conservation campaigns, policy development, partnerships, and research as well as the study shed. These items indicate that the solar facility sits within a larger sustainability agenda linking renewable energy, operational efficiency, and energy literacy.

A chronology distinction is important: the institutional SDG 7 PDF groups the study shed within its portfolio summary, but the public news article documents the formal turnover in June 2025. This report therefore treats the April 2024 Greenergy meeting as the partnership-exploration milestone and the June 2025 turnover as the dated completion milestone. The distribution chart is presented as the institutional report's own contribution profile, not as a strict list of only assets installed during calendar year 2024.

Comparative Progression Across 2024 and 2025

Documented sequence: from partnership exploration to a delivered solar learning facility



**The ocular visit is described in the supplied 2025 narrative; the linked page was not independently accessible during preparation.*

Figure 2. Timeline of key milestones in the transition from partnership exploration to the solar-powered study shed. The March 2025 ocular visit is based on the supplied narrative.

Interpretation: The chronology shows a sequence rather than a direct numerical year-on-year comparison. In April 2024, CCCDO initiated discussion with Greenergy Solar PH about potential sustainability collaboration. In March 2025, the supplied narrative records an ocular visit to assess solar opportunities at the Agusan Campus. By June 2025, CCCDO had received a solar-powered study shed through Mindanao Goes Solar. Taken together, the sequence is consistent with a progression from relationship-building to site assessment and then to a visible installation for student use.

The significance of the progression lies in the change in implementation form. The 2024 report mainly describes an institutional framework—policy, monitoring, awareness, partnerships, and research. The 2025 study shed provides a tangible renewable-energy learning space. It represents a move from broad strategic alignment toward a completed campus-facing facility, although the sources do not establish the extent to which this facility changed the College's overall electricity demand.

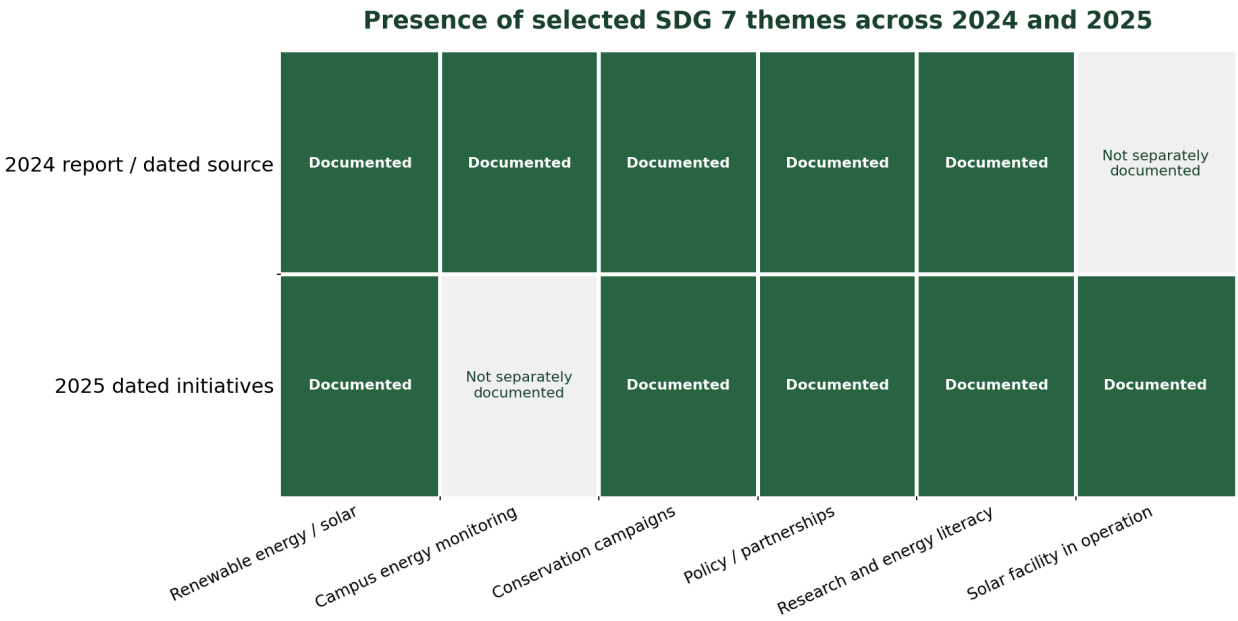


Figure 3. Presence of selected themes in the 2024 report and 2025 dated narrative. "Not separately documented" does not mean that a related activity was absent; it means a distinct year-specific result was not identified in the reviewed source.

Interpretation: Energy conservation, partnerships, and education are present in the documented 2024 portfolio, while the 2025 record adds a clearly dated solar facility and a solar-site assessment. This suggests a shift in the visibility of implementation. The 2024 energy-monitoring system and conservation campaigns are described in the institutional progress PDF, but the reviewed 2025 narrative does not supply a comparable 2025 consumption figure or a new building-level monitoring total. The visual therefore compares what is explicitly documented rather than implying that an activity stopped or was newly introduced solely because it is not separately listed in one year's narrative.

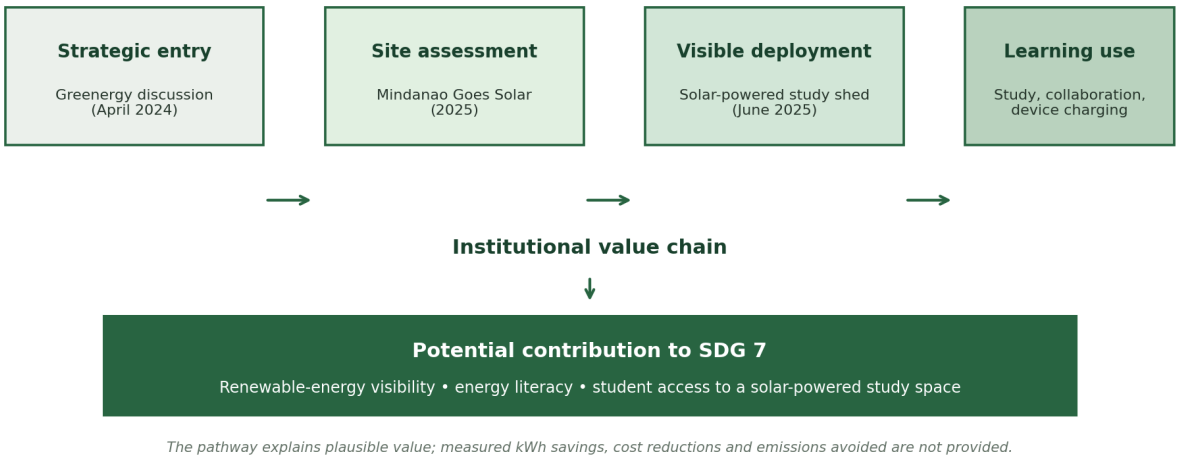


Figure 4. Conceptual pathway explaining how the 2024–2025 initiatives may contribute to SDG 7. This is an analytical model, not a quantified impact claim.

Interpretation: Partnership exploration can lead to technical assessment; assessment can inform site decisions; and a delivered solar-powered study shed can provide a real-world setting for learning and device use. Alongside the 2024 policy, monitoring, and awareness strands, these stages show how institution-level sustainability can connect operational management and student experience. The pathway is plausible and consistent with the activities described, but the source material does not provide generation, savings, or emissions data to quantify the ultimate results.

Comparative Analysis by Dimension

Dimension	2024 baseline	2025 progress documented	Why the change matters
Renewable energy adoption	Exploratory meeting with Greenergy Solar PH; institutional report categorizes renewable energy as the largest share of its SDG 7 contribution distribution.	Mindanao Goes Solar ocular visit described in the supplied narrative; solar-powered study shed formally turned over June 2, 2025.	The record moves from exploring possible partnerships toward a completed renewable-energy facility available for student use.
Energy efficiency and monitoring	ICT energy/water dashboard described for three buildings; report names conservation measures and campaigns.	The supplied 2025 narrative frames energy saving and monitoring as part of ongoing SDG 7 reporting but provides no comparable 2025 usage series.	Monitoring and conservation provide the operational context around renewable energy, although changes in demand are not quantified.
Campus awareness and energy literacy	Switch-Off Drive, Green Classroom Challenge, and Eco-Literacy Month; source reports 2,000 students and faculty reached.	The study shed provides a visible applied learning site; curriculum integration and sustainability education remain part of the broader portfolio described.	Connecting infrastructure with awareness can make clean energy understandable through practical student experience.
Partnerships and institutional coordination	Greenergy Solar PH exploratory discussion; report also names City Engineering Office, COWD, DOE-10, and four inter-agency collaborations.	Mindanao Goes Solar delivered the study shed; a donation deed was signed in June 2025; an ocular assessment was described.	Partnerships connect institutional priorities with external expertise, assessment capacity, and implementation resources.
Research and curriculum	Report describes five research papers in its category table, energy-related	The study shed is described as an applied learning space; separate 2025	Infrastructure can support experiential energy literacy, but documented learning

Dimension	2024 baseline	2025 progress documented	Why the change matters
	research topics, and integration in STEM/General Education.	student research outputs or curricular outcome statistics are not itemized in the supplied narrative.	outcomes should not be inferred from installation alone.
Measured energy outcomes	2024 source gives selected counts and shares but no complete baseline series of kWh use, generation, or costs.	No matched data on kWh generated, bill savings, avoided emissions, system uptime, or comparable energy-use changes.	The comparison establishes activity and implementation progression, not a percentage reduction in energy costs or carbon emissions.

Alignment with SDG 7

CCCDO's documented initiatives relate to several SDG 7 themes. The solar-powered study shed most visibly connects to the adoption and demonstration of renewable energy, while the monitoring dashboard and conservation campaigns relate to energy efficiency and responsible consumption. The Greenergy Solar PH discussion and Mindanao Goes Solar engagement show the importance of partnerships in exploring and implementing energy technologies. The report's research and curriculum component connects sustainable energy with teaching and student knowledge.

SDG 7 theme	Documented CCCDO contribution	Interpretation
Affordable, reliable, modern energy (Target 7.1)	Solar-powered study shed turned over in June 2025; source describes student study, collaboration, and device-powering uses.	Adds a solar-powered amenity to the campus environment; the supplied data do not quantify its availability or effect on overall campus power reliability.
Renewable energy (Target 7.2)	Greenergy Solar PH exploratory discussion in 2024; solar-site assessment described in 2025; study shed delivered in June 2025.	Shows progression from partnership exploration to a tangible solar application. System capacity and generation figures are not supplied.
Energy efficiency (Target 7.3)	Digital monitoring across three buildings as described in the SDG 7 report; Switch-Off Drive and Green Classroom Challenge; localized energy-efficiency guidance.	Combines technology, policy, and behavior. A measured reduction in total consumption is not included in the reviewed figures.

SDG 7 theme	Documented CCCDO contribution	Interpretation
Energy infrastructure and technology (Target 7.b)	Campus-facing solar study shed and external cooperation with clean-energy partners.	Provides a visible example of clean-energy technology in an educational setting, while institution-wide scale is not quantified.
Education and partnerships as enabling dimensions	Research outputs, energy literacy within courses, and collaboration with external agencies and providers.	Supports knowledge and implementation capacity that can complement physical infrastructure.

Conclusion

The 2024–2025 comparison presents CCCDO's SDG 7 work as a progression from a broad institutional framework toward a more visible application of solar energy in student learning space. In 2024, the College's documented portfolio emphasized renewable-energy planning, energy and water monitoring, conservation campaigns, institutional policy, inter-agency cooperation, research, and curriculum integration. The April 23 exploratory meeting with Greenergy Solar PH established a dated point of engagement with an energy provider and positioned clean energy within the College's sustainability discussions.

In 2025, the ocular visit described in the supplied narrative and the formal turnover of the solar-powered study shed on June 2 demonstrate a more concrete stage of activity. The shed provides a practical setting for students to study, collaborate, and power devices using a solar-powered facility. This transition matters because it connects the College's sustainability commitments with a visible campus application and an opportunity for energy literacy through everyday use.

The evidence supports describing the 2025 milestone as implementation progress and a broadening of the visible clean-energy portfolio. It does not support a numerical claim that overall electricity consumption or costs fell by a particular amount, that carbon emissions were reduced by a given percentage, or that the campus became energy independent. The report's strongest conclusion is therefore a documented institutional progression: partnership exploration and sustainability planning in 2024, followed by site assessment and a completed solar-powered study facility in 2025, supported by the wider policy, monitoring, conservation, partnership, and education framework described by CCCDO.

Source and Verification

[1] City College of Cagayan de Oro. (2024, May 6). Synergy in Motion: Greenergy Solar Ph and City College of Cagayan de Oro. https://citycollegecdo.edu.ph/article/757

[2] City College of Cagayan de Oro. (2025, March 19). Mindanao Goes Solar Conducts Ocular Visit at CCCDO Agusan Campus. https://citycollegecdo.edu.ph/sdg-7

[3] City College of Cagayan de Oro. (2025, July 17). CCCDO Receives Solar-Powered Study Shed from Mindanao Goes Solar. https://citycollegecdo.edu.ph/article/1049

[4] City College of Cagayan de Oro. (n.d.). *Progress Report on SDG 7: Affordable and Clean Energy [Institutional progress report supplied for this analysis].

https://citycollegecdo.edu.ph/dist/SDG/SDG7-Affordable-and-Clean-Energy-Progress-report.pdf